

INVITED TALK / ABSTRACT

Band Width Estimates and Rigidity of Manifolds with Negative Curvature

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Abstract

Motivated by a recent width estimate of Zhou for metrics with positive sectional curvature on $T^2 \times [-1, 1]$, we investigate width-type inequalities under lower Ricci curvature bounds. Our main tool is Gromov's μ -bubble technique, based on minimizers of weighted area functionals, which has become a powerful method in geometric analysis. We establish optimal Lipschitz lower bounds for a class of proper smooth functions on connected, orientable, noncompact three-dimensional manifolds with $\text{Ric} \geq -2$. As applications, we derive width estimates for manifolds of the form $\Sigma^2 \times [-1, 1]$, where Σ is either a torus or a hyperbolic surface. In the hyperbolic case, the boundary area plays a fundamental role, revealing a phenomenon absent in the nonnegative curvature setting.

Keywords: band width estimates; rigidity; negative curvature; lower Ricci curvature bounds; Gromov's μ -bubble technique; weighted area functionals; Lipschitz lower bounds; hyperbolic surface